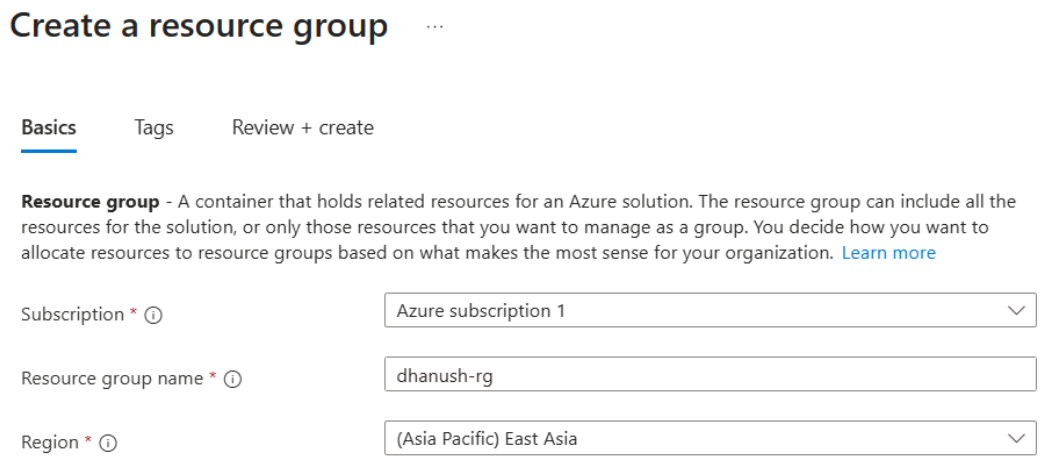
# LAB 3 : Create 3 Linux VMs in Availability Zones

## STEP 1 : Resource Group

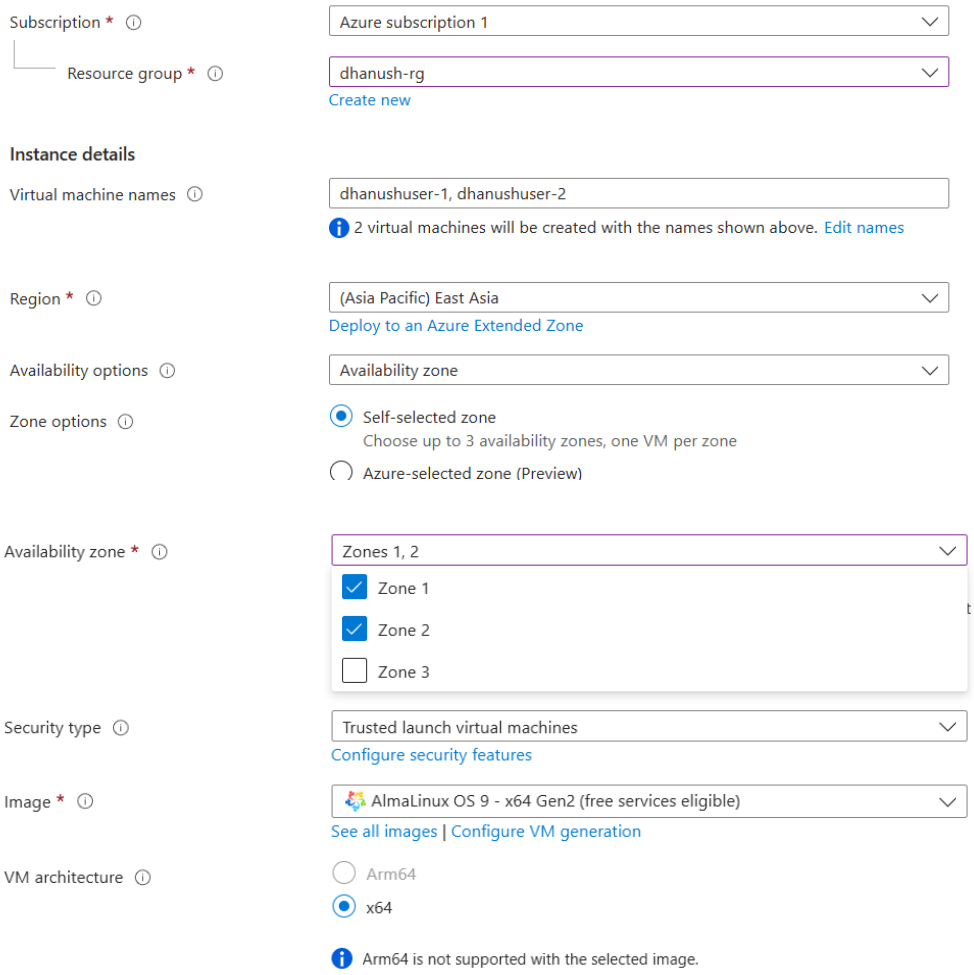
- **Name:** <studentname>-rg




## STEP 2 : Create Resource Group

Go to Resource groups → Create.

- Resource group name: <studentname>-rg
- Region: Choose a region that supports Availability Zones (e.g., *East US 2*, *West Europe*, *Central India*).




- Click Review + Create → Create.

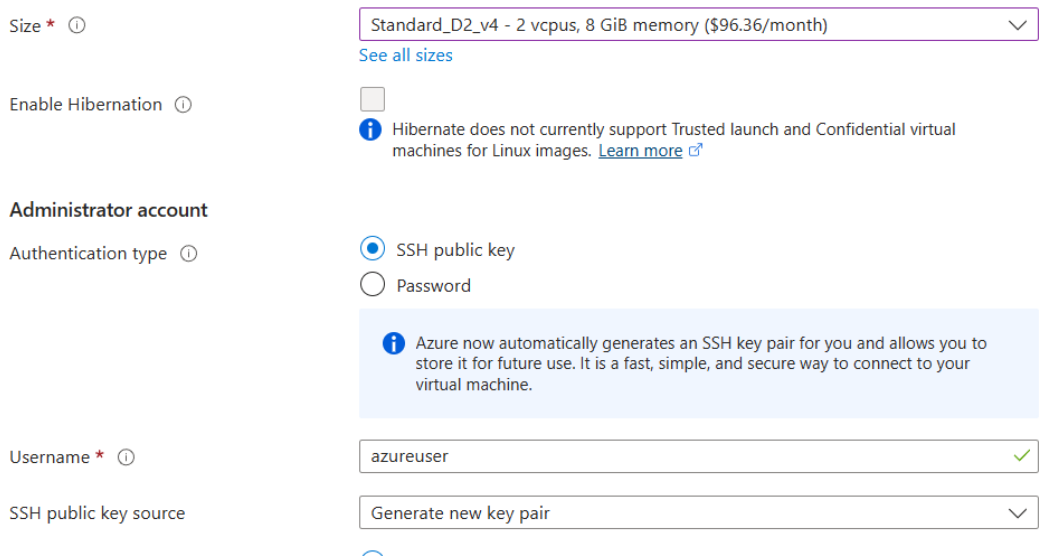
## Create 2 (or) 3 Linux VMs across Zones

We'll create VMs one by one, placing each in a different Availability Zone.

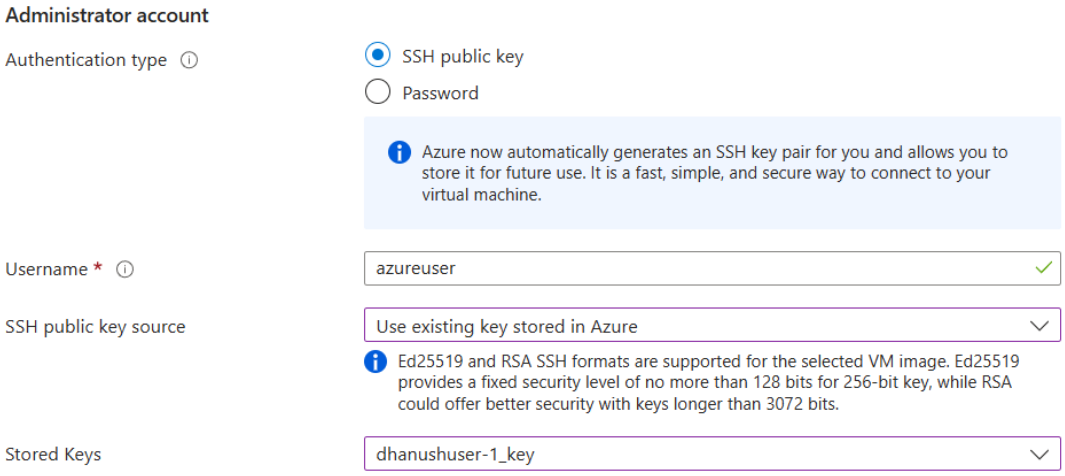
## VM1

- Resource group: <studentname>-rg
- VM name: <studentname>-linux-vm1
- Region: Same as RG
- Availability options: Availability zone

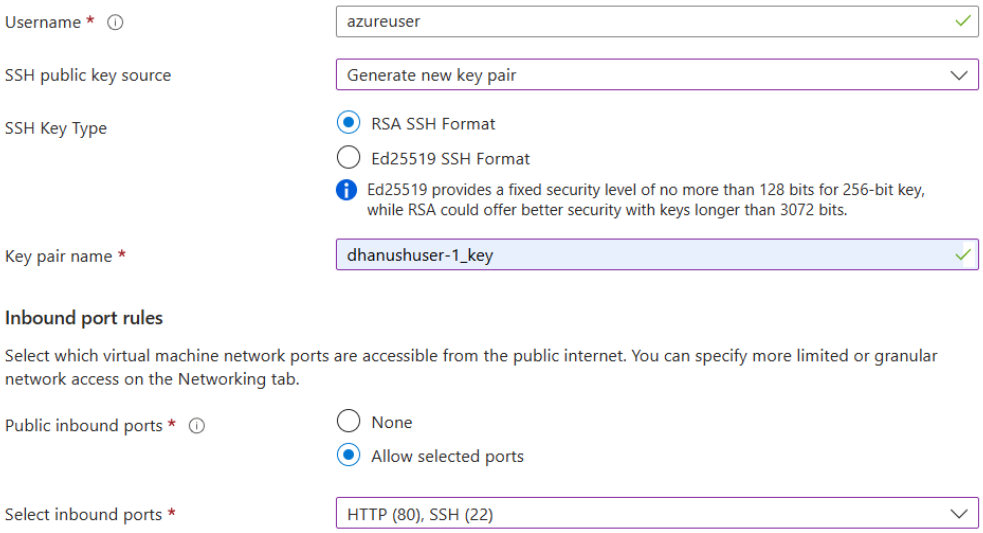
- Zone: 1
- Image: AlmaLinux OS 9 – x64 Gen2
- Size: Standard_B1s



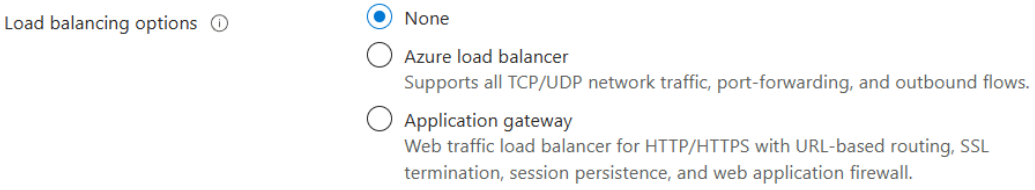

- Authentication: SSH public key (generate or use existing)



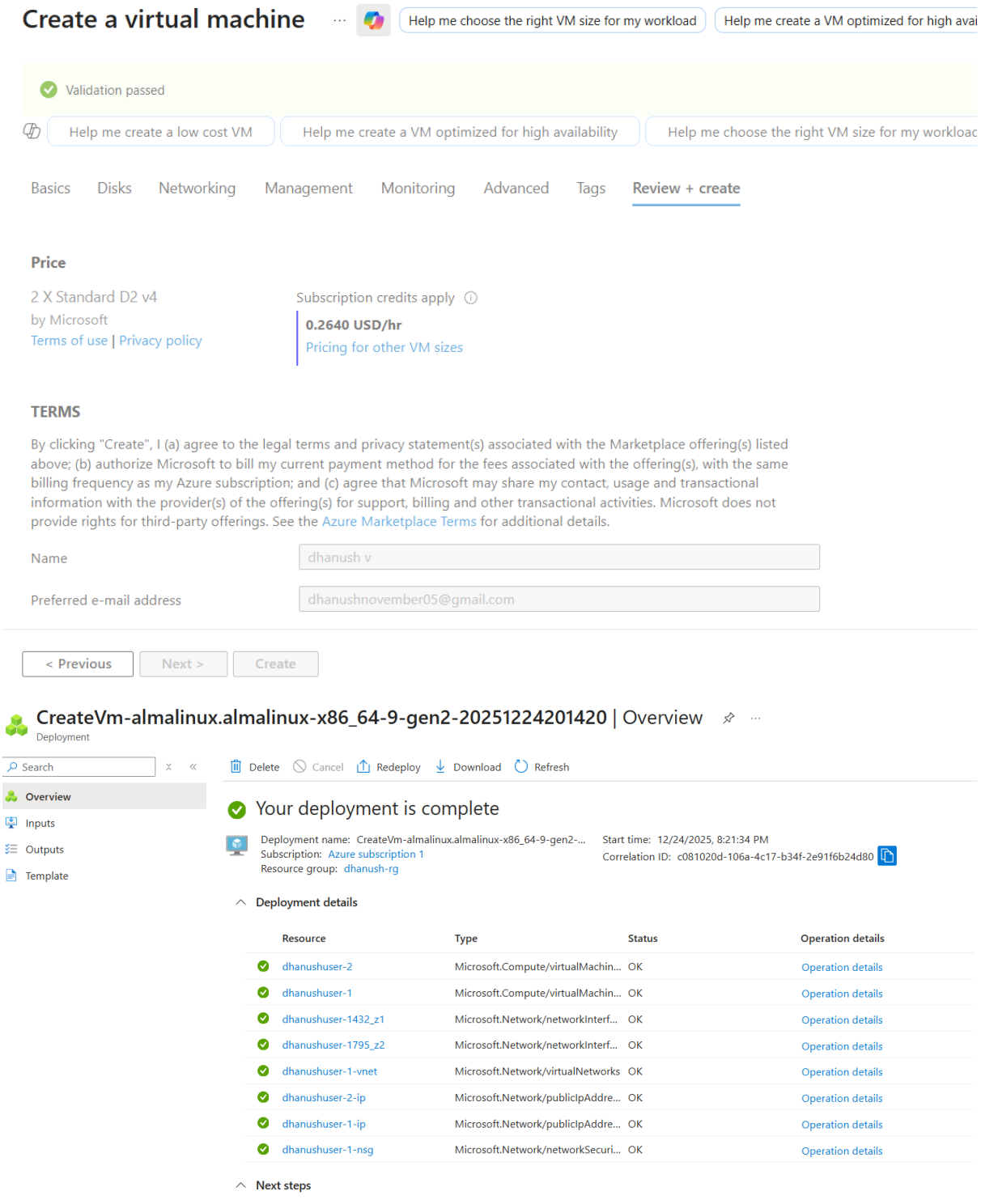

- Inbound ports: Allow SSH (22)




Before review + create go to the networking and select load balancing options (None)




Click Review + Create → Create




# VM2

- VM name: <studentname>-linux-vm2
- Zone: 2

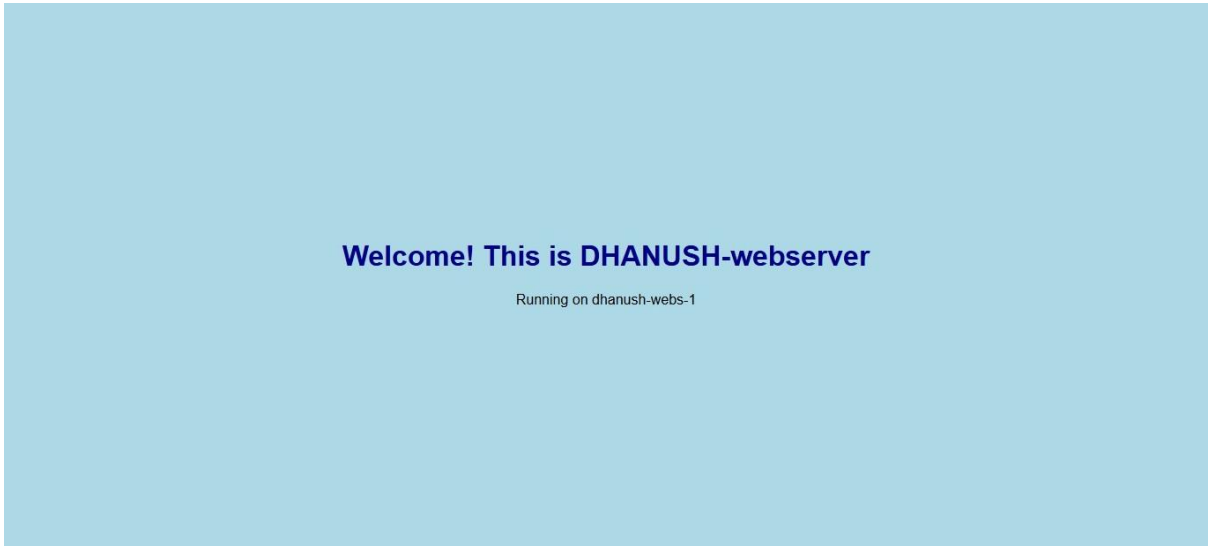
- Other settings same as above.

VM3

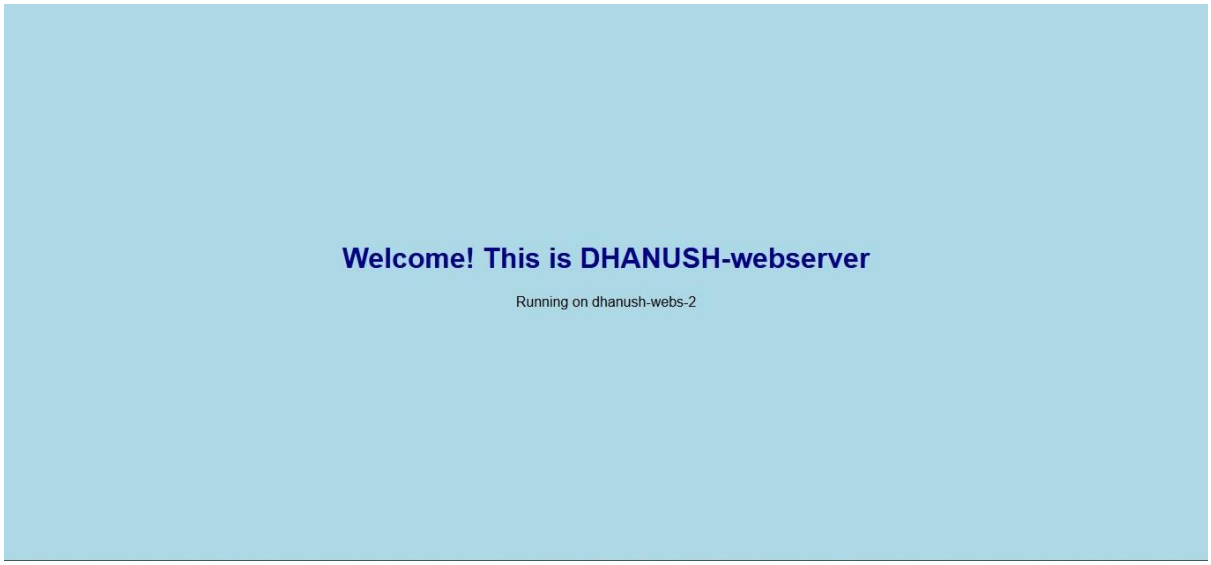
- VM name: <studentname>-linux-vm3
- Zone: 3
- Other settings same as above.

STEP 3 : Copy the public ip address for the zones and paste it in the browser for the verification

Zone 1




Zone 2




## For Verification

1. After deployment, check all 3 VMs:
    - Go to Virtual machines → Select a VM → Overview → confirm Availability Zone.
2. Connect to each VM:

- ssh -i <studentname>-key.pem azureuser@<vm-public-ip>
- hostname && uptime

3. Confirm each VM is in a different zone (1, 2, 3).

# Outcome

- You now have **3 Linux VMs**, each in a different **Availability Zone**, ensuring resilience against a **datacenter failure** within the region.
- If one datacenter goes down, the other two VMs will still be running.